Practical #6

Practical Title:
To deploy and manage a simple web application using version control and hosting techniques

Theory:

1. What is Web Hosting?

Web hosting is a service that allows individuals or organizations to **store their website files on a server** and make them accessible on the internet. It ensures that the website is available to users anytime.

2. What is GitHub Hosting (GitHub Pages)?

GitHub Pages is a free hosting service provided by GitHub that allows users to **deploy static websites directly from a repository**. It is commonly used for portfolios, projects, and simple web apps.

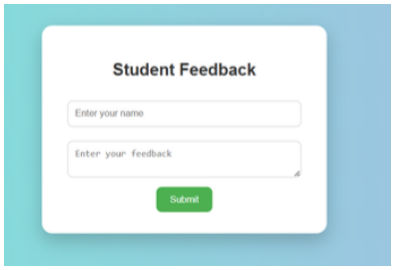
3. Role of Version Control in Deployment:

Version control helps in managing and tracking code changes during deployment. It allows:

- Easy updates and rollback
- Collaboration among developers
- Maintaining history of changes
- Automated deployment through integration tools

Procedure:

Step 1: Create a simple web application (index.html)



Step 2: Initialize Git repository

```
ACER@DESKTOP-8J9C8K7 MINGW64 ~/devops-practical5/devops-practical6 (main)
$ git init
Initialized empty Git repository in C:/Users/ACER/devops-practical5/devops-practical6/.git/
```

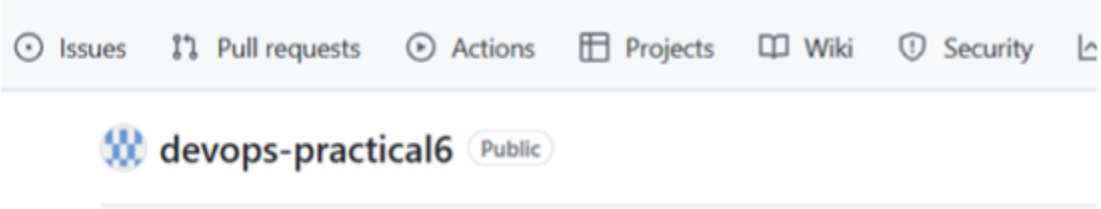
Step 3: Add and Commit files

```
ACER@DESKTOP-8J9C8K7 MINGW64 ~/devops-practical5/devops-practical6 (maste
$ git add .
git commit -m "Initial web app"
On branch master

Initial commit

nothing to commit (create/copy files and use "git add" to track)
```

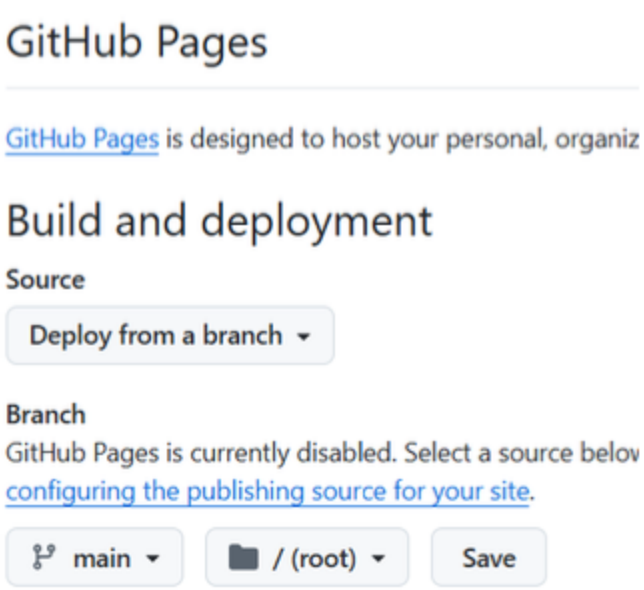
Step 4: Create GitHub repository



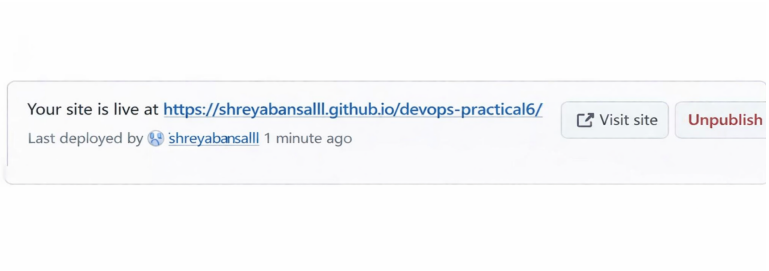
Step 5: Push code to GitHub

```
ACER@DESKTOP-819C8K7 MINGW64 ~/devops-practical5/devops-practical6
$ git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/22, done   | 283.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ishanjain/devops-practical6.git
* [new branch]   main -> main
branch 'main' set up to track 'origin/main'.
```

Step 6: Enable GitHub Pages



Step 7: Access deployed website using provided link



Activity / Task:

Task 1:
Deploy your own web application using GitHub Pages.

Repository Link:
https://github.com/shreyabansalll/devops-practical6

Live Website Link:
https://shreyabansalll.github.io/devops-practical6/

Task 2:
Make changes in your code and update the live website.

Commands Used:

git add .

git commit -m "Updated website"

git push origin main

Code Blame

```
Hello DevOps
<p>Updated website</p>
```

Commands Used:

```
git add .
git commit -m "Updated website"
git push
```

Task 3:
Explain the benefits of using GitHub for deployment.

Benefits of using GitHub for deployment:

- Free hosting for static websites
- Easy integration with Git version control
- Automatic deployment on push
- Supports collaboration
- Provides version history and rollback

Advantages:

1. Free and easy hosting
2. Quick deployment process
3. Version control integration
4. Accessible from anywhere
5. Supports continuous updates

Challenges:

1. Only supports static websites
2. Limited customization options
3. Requires basic Git knowledge